

PULSED EMITTER – 20 kHz / 5 μ s / ~500 mA

Emitter loop is local: C20 supplies the 5 μ s / 500 mA pulse so the cable carries only the ~50 mA average. R19 damps the gate edge; R20 holds Q1 off while the main board is unpowered or the cable is unplugged.

TRANSIMPEDANCE + x12 AC GAIN

The circuit diagram illustrates a transimpedance amplifier (TIA) with a gain of 12. It is composed of two operational amplifiers, U8A and U8B, both TLV9062IDR.

U8A (TLV9062IDR) - Transimpedance Amplifier Stage:

- The non-inverting input (OPT_VRE6) is connected to ground.
- The inverting input (PD_TIA_IN) is connected to a photodiode (PD1, BPW34S) and a feedback network consisting of a capacitor C21 (10pF) and a resistor R21 (10k) in parallel.
- The output of U8A (TIA_OUT) is connected to the non-inverting input of U8B.

U8B (TLV9062IDR) - AC Gain Stage:

- The inverting input (AMP_INV) is connected to ground through a resistor R22 (10k).
- The feedback path consists of a capacitor C22 (10nF) and a resistor R23 (120k) in parallel.
- The output of U8B (OPT_AMP) is the final output.

The circuit is powered by a +3V3 supply.

The diagram illustrates a threshold comparator circuit with hysteresis, implemented using a TS3021N VLT op-amp (U4). The circuit is powered by a +3V3 supply and includes a 100nF decoupling capacitor (C25).

Key Components and Connections:

- Op-Amp (U4):** TS3021N VLT. The non-inverting input (pin 1) is connected to the output (pin 1) via a feedback resistor R28 (470k), creating positive feedback (hysteresis). The inverting input (pin 2) is connected to the input signal CMP_THRESHOLD via a resistor R27 (10k). The output (pin 1) is also connected to the input signal CMP_THRESHOLD via a resistor R26 (22k), creating negative feedback.
- Power Supply:** The op-amp is powered by a +3V3 supply. A 100nF capacitor (C25) is connected across the supply pins to decouple the power.
- Output:** The op-amp output (pin 1) is labeled CMP_OUT. It is connected to the input signal CMP_THRESHOLD via a resistor R32 (33R).

Signal Flow:

- The input signal **CMP_THRESHOLD** is applied to the inverting input (pin 2) through resistor R27.
- The output of the op-amp, **CMP_OUT**, is connected to the input signal **CMP_THRESHOLD** through resistor R32.

1.65 V REFERENCE

CABLE TO MAIN BOARD

J5

TO MAIN BOARD

7 +3V3

6 OK_COMP

5 OK_LED_EN

4 OK_LED_EN

3 OK_LED_EN

2 SYS_SW

1 SYS_SW

+3V3

SYS_SW

BalancerREF_OptHead – optical tach head, at 90 degrees to the main PCB on the aluminium L-bracket. One pivot bolt aims the emitter; add a second fastener (arc slot or serrated lock nut) or it drifts under vibration and loses RPM lock. The baffle between D1 and PD1 and the red acrylic window belong to this board. Designators match BalancerREF Rev C so the existing pin audits stay valid.

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